TELE9782 Presentation Script

17 November 2026 | Hanqing Zhao & Ning Mao | Main talk 10:00

Hanqing 00:00–05:00 (601 words) · Ning 05:00–10:00 (570 words). Counts include spoken transitions. Pause while pointing at figures. Appendix responses are outside the timer.

01 Design and Simulation of a D-Band Microstrip Patch Antenna

Hanqing Zhao | 00:00–00:30 | 30 seconds

Good morning. We are Hanqing Zhao and Ning Mao. For TELE9782, we designed an inset-fed microstrip patch using CST Studio Suite. The aim was to exceed 6.5 dBi of directivity in the specified D-band range. We will explain how the starting dimensions were calculated, how the model was tuned, and how we checked the final result.

Transition — First, here are the requirements that defined the design.

Pronunciation — D-band · dee band; dBi · dee-bee-eye

02 Project requirements

Hanqing Zhao | 00:30–01:20 | 50 seconds

The specified frequency range is 130 to 170 gigahertz. We chose 150 gigahertz, near the middle of that range, as the nominal design point. The main target is maximum directivity greater than 6.5 dBi. The brief also asks us to compare the analytical resonance with simulation, plot S eleven and extract its minus-ten-decibel bandwidth, report directivity and realized gain versus frequency, show radiation efficiency, and obtain the E-plane and H-plane patterns. We use the full sweep to see where the antenna responds. The impedance bandwidth is a separate quantity defined by the two minus-ten-decibel crossings.

Transition — The rectangular patch model gives us a starting size for that frequency.

Pronunciation — GHz · gigahertz; S_{11} · S eleven; E-plane · ee plane

03 Rectangular patch theory

Hanqing Zhao | 01:20–02:30 | 70 seconds

In the fundamental T M one zero mode, one half-wave variation fits along the resonant length. The fringing fields at the two radiating edges act like two slots. Their broadside fields add, giving a main lobe normal to the patch. The wave travels through fields in both the substrate and the air, so we use an effective relative permittivity. This makes the guided wavelength shorter than the free-space wavelength. Fringing makes the effective patch length longer than the metal length, so we subtract the edge extension twice to get the physical length. Directivity describes how much radiation is concentrated in one direction. It compares the peak radiation intensity with the average over the sphere. The patch equations give a starting geometry, and the full-wave farfield gives the actual directivity of the model.

Transition — Here are the dimensions that come out of those equations.

Pronunciation — TM_{10} · tee-em-one-zero; fringing · FRIN-jing; permittivity · per-mih-TIV-ih-tee

Hanqing Zhao · continued

04 Analytical design and final parameters

Hanqing Zhao | 02:30–03:45 | 75 seconds

At 150 gigahertz, the free-space wavelength is almost two millimetres. For a forty-micrometre polyimide substrate with relative permittivity 3.5, the starting width is 0.666 millimetres. The effective permittivity is about 3.203. The half guided wavelength is 0.558 millimetres, and the fringing correction is 0.019 millimetres at each edge. That gives a starting length of 0.520 millimetres. After tuning, we kept the width at 0.667 millimetres and shortened the length to 0.513. Length primarily sets resonance. Width influences the radiating aperture and input impedance. Substrate thickness changes the field distribution and the match. Feed position controls how the line couples to the patch. Applying the same formula to the final dimensions gives 152.00 gigahertz. The simulated S eleven minimum is at 148.84, about 2.08 percent lower. The finite ground and feed affect the input response.

Transition — We then modelled those features explicitly in CST.

Pronunciation — Polyimide · pol-ee-IM-ide; micrometre · MY-kroh-mee-ter

05 CST model and simulation setup

Hanqing Zhao | 03:45–05:00 | 75 seconds

This is the final CST model. The gold patch and feed sit on polyimide, above a finite gold ground. A uniform launch extends from the board to the port. The waveguide port excites one quasi-TEM microstrip mode towards the patch, with a 50-ohm reference for the S parameters. The time-domain solver sweeps 130 to 170 gigahertz. Open boundaries approximate free space, and the model includes dielectric loss and finite gold conductivity. The mesh resolves the thin substrate, the feed gaps and the port region. Thirteen farfield monitors sample the band, including points around the matched region and at exactly 150 gigahertz. We also compare two meshes and check the power balance and the full-sphere directivity normalization. Those checks matter because a completed solver run can still produce an unreliable farfield.

Transition — Ning will explain how the model was corrected and how the final results were verified.

Pronunciation — quasi-TEM · KWAY-z eye tee-ee-em; hexahedral · hex-uh-HEE-dral

Ning Mao · results and validation

06 Troubleshooting and optimisation

Ning Mao | 05:00–06:10 | 70 seconds

Thanks, Hanqing. The original run displayed a maximum directivity near minus 20.92 dBi. That was physically inconsistent for a normalized full-sphere maximum. The port faced away from the patch, and changing its orientation exposed an insufficient uniform feed length. We rebuilt the launch and parameterised the ground. The reconstructed model ran, with a baseline S eleven minimum near minus 10.18 dB. The inset sweep improved the match. The 0.17 millimetre trial reached minus 14.18 dB, and the 0.19 trial reached minus 26.82. We shortened the patch to move the matched region upwards. A later farfield check also exposed an error with the near-model integration surface. The final configuration uses a global enclosing surface and a stricter decay criterion. The port and farfield changes made the calculation credible. Inset tuning primarily improved the match.

Transition — Here is the final S eleven response after those changes.

Pronunciation — inset · IN-set; normalization · nor-muh-lih-ZAY-shun

07 Resonance and S_{11}

Ning Mao | 06:10–07:15 | 65 seconds

The orange marker shows 150 gigahertz. S eleven there is minus 13.424 dB, so about 4.55 percent of incident power is reflected. The deepest sampled dip is minus 38.859 dB at 148.840 gigahertz. We call it the match minimum because the input reflection also depends on the feed and impedance reference. The minus-ten-decibel crossings are 147.106 and 150.619 gigahertz. Subtracting the two gives a bandwidth of 3.513 gigahertz. Dividing by their midpoint gives 2.36 percent. The selected 150 gigahertz point is within that interval and has a dedicated farfield monitor. This lets us compare input matching and radiation at exactly the same frequency.

Transition — The next slide shows what the antenna does with the power it accepts.

Pronunciation — fractional bandwidth · FRAK-shuh-nuhl BAND-width; impedance · im-PEE-dance

08 Directivity and realized gain

Ning Mao | 07:15–08:25 | 70 seconds

The left plot shows maximum directivity and the right plot shows maximum realized gain at the monitor frequencies. At 150 gigahertz, directivity is 7.460 dBi. The requirement is greater than 6.5 dBi, so the margin is 0.960 dB at a matched frequency. Directivity tells us where the radiated power goes. Realized gain also accounts for losses in the antenna and reflection at the input. The realized gain at the same point is 5.804 dBi. We can reproduce it by adding ten times the logarithm of total efficiency to the directivity in dBi. The largest sampled directivity is at 154 gigahertz, where the input match is poor. We report the result at 150 because it demonstrates useful matching and directivity together.

Transition — The efficiency and the principal-plane cuts explain the radiation in more detail.

Pronunciation — realized · REE-uh-lyzd; isotropic · eye-soh-TROP-ik

Ning Mao · continued

09 Efficiency and radiation patterns

Ning Mao | 08:25–09:30 | 65 seconds

The first graph shows radiation efficiency over frequency. At 150 gigahertz, 71.55 percent of the accepted power is radiated. If we also include the reflected power, total efficiency is 68.30 percent. The two cuts show the directional pattern at 150 gigahertz. The broadside electric field is mainly y-polarized. This makes the y z cut the E-plane and the x z cut the H-plane. Both show a forward lobe. In the full three-dimensional pattern the maximum is close to positive z, about four degrees from broadside on a two-degree sampling grid. The cuts give the shape of the beam in two perpendicular planes, while the full sphere supplies the maximum directivity.

Transition — We close with the checks that support the reported performance.

Pronunciation — broadside · BROAD-side; polarisation · poh-luh-rih-ZAY-shun

10 Validation and conclusion

Ning Mao | 09:30–10:00 | 30 seconds

Between the two meshes, directivity changed by only 0.037 dB at 150 gigahertz. The spherical mean was 0.9991, and the power-balance residual was 0.754 percent. The saved project reopened and reran successfully. At 150 gigahertz, this single patch delivers 7.460 dBi of simulated directivity with S eleven below minus 10 dB.

Transition — Thank you. We are happy to take questions.

Pronunciation — residual · ree-ZIJ-oo-uhl

Appendix speaking notes · Q&A only

11 Transmission-line calculation

Hanqing Zhao | 20–30 seconds if asked

The effective length is about 0.5584 millimetres. Fringing adds around 0.019 millimetres at each edge, so the starting metal length is 0.5204 millimetres. Replacing the starting dimensions with the final ones gives the 152 gigahertz estimate.

Transition — Appendix B shows which parameters were actually entered.

Pronunciation — Leff · L effective; ΔL · delta L

12 Final geometry and materials

Hanqing Zhao | 20–30 seconds if asked

The patch is 0.667 by 0.513 millimetres on forty micrometres of polyimide. The ground and launch use the same finite gold thickness. These are nominal D-band material assumptions.

Transition — Appendix C shows the feed cross-section and its reference impedance.

Pronunciation — tanδ · tan delta; σ · sigma

13 Waveguide port and inset matching

Hanqing Zhao | 20–30 seconds if asked

The waveguide port sits at y equals minus 1.2 millimetres. A uniform launch supports the microstrip mode, whose computed line impedance is 49.47 ohms at 150 gigahertz. We use a 50-ohm S-parameter reference.

Transition — Appendix D shows how the accepted result was checked.

Pronunciation — quasi-TEM · KWAY-z eye tee-ee-em

14 Mesh, power and rerun checks

Ning Mao | 20–30 seconds if asked

The two meshes use identical geometry and farfield settings. The directivity difference is 0.037 dB. The power left after subtracting radiation and material losses from accepted power is 0.754 percent of accepted power.

Transition — Appendix E gives the frequency results used in the plots.

Pronunciation — spherical · SFEH-rih-kuhl

Appendix speaking notes · continued

15 Frequency results at all 13 monitors

Ning Mao | 20–30 seconds if asked

Each row corresponds to an actual monitor. At 149 gigahertz the sampled realized gain peaks. At 154 gigahertz directivity peaks, but the match is poorer. The 150 gigahertz row gives the operating point.

Transition — Appendix F shows the full three-dimensional pattern.

Pronunciation — realized · REE-uh-lyzd

16 Polarisation and 3D farfield

Ning Mao | 20–30 seconds if asked

The electric field near broadside is y-polarized, which determines the E-plane. The full-sphere maximum is near positive z. The angular sampling is two degrees, so we report the beam angle approximately.

Transition — Appendix G gives the diagnostic history and references.

Pronunciation — theta · THAY-tuh; phi · fie

17 Diagnostic history and references

Ning Mao | 20–30 seconds if asked

We rejected the original negative maximum and the later near-model normalization error. The accepted calculation uses the enclosing surface and passes the spherical check. The analytical reference is the standard microstrip transmission-line model.

Transition — Return to the plot or result that addresses the next question.

Pronunciation — normalization · nor-muh-lih-ZAY-shun